

HAT-P-36b

R-0711

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260323 · created 2026-07-27T22:48:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461122.784 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.015
Transit depth (Rp/Rs)^2	1.97 +/- 0.42 [%]
Orbital Inclination (inc)	81.42 +/- 2.44
Airmass coefficient 1 (a1)	1.01 +/- 0.017
Airmass coefficient 2 (a2)	-0.007 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.67 %
Best Comparison Star	#2 - [138, 397]
Optimal Aperture	3.9
Optimal Annulus	13.77
Transit Duration (day)	0.079 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.015 vs 0.126 (deviation 0.9 σ)
Duration fit vs published	0.079 d vs 0.0925 d (deviation 0.87 σ)
Mid-time O–C	1.3 min
Depth SNR	8.5
Residual scatter	1.67 %

Light curve

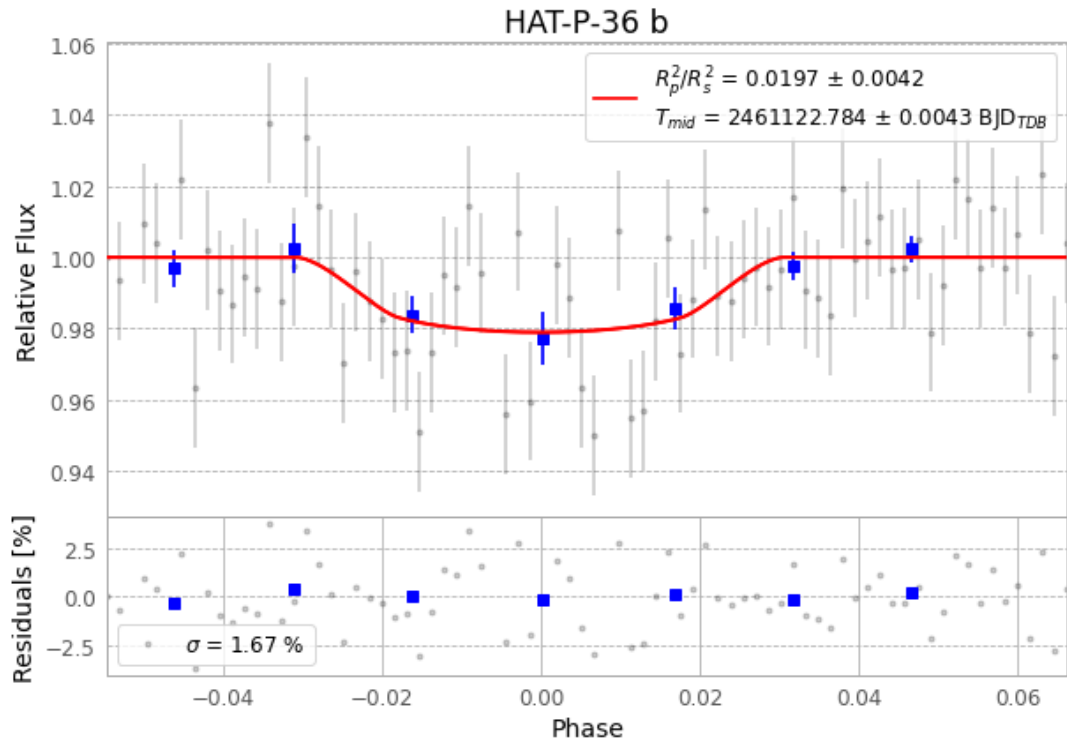


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-03-22.png (SHA-256 5aacff277ae59c16...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01fe3599d9e4debe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f7446be

Data provenance

78 FITS frames (51 MB), HATP-36260323045715.FITS → HATP-36260323084816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260323.log (d38a39dd9f07...), FinalLightCurve_HAT-P-36 b_2026-03-22.png (5aacff277ae5...), FinalLightCurve_HAT-P-36 b_2026-03-22.csv (fa5ac37df0ed...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)